

Osaid Khan

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EDUCATION

Pace University

Master of Science (MS) in Computer Science (GPA - 4.0/4.0)

New York City, NY

Jan 2024 - Dec 2025

Dr. A.P.J. Abdul Kalam Technical University

Bachelor of Technology (B.Tech) in Computer Science

Noida, India

Jul 2016 - Sep 2020

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, Rust

Frameworks: Spring Boot, React, FastAPI, FastMCP, LangGraph, LangChain, LangSmith, PyTorch

Others: Git, CI/CD, Google Cloud, Node.js, Kafka

Databases: PostgreSQL, MongoDB, Redis

AWS Services: EC2, RDS, S3, Lambda, CloudWatch, Step Functions, CloudFormation, CloudFront, API Gateway

AI: Claude code, Codex, Pi coding-agent

PROFESSIONAL EXPERIENCE

Software Engineer

Phoenix, AZ

Sperse

Feb 2026 - present

- Architected and delivered an **MCP interaction layer** using **LangChain** and **LangGraph**, translating natural-language intents into **validated API calls** across **1000+ platform endpoints**, functioning as a high-throughput **data access tool** for **massive scale** operations
- Collaborated with product managers to architect **generative code solutions** (API-aware templates and SDK patterns), accelerating the **software development lifecycle (SDLC)** and ensuring the long-term maintainability of core **API stacks**
- Established a robust **observability layer** using **Langfuse** and **LangSmith** for **tracing** and **structured logs**, enabling the identification of software improvements and ensuring best-in-class performance of **distributed services**
- Optimized **agent concurrency** and prompt routing to maximize **KV-cache hit rate**, reducing end-to-end latency by **60%** and lowering operational costs through rigorous **algorithm** and **data structure** optimization

AI Researcher

New York City, NY

Pace Artificial Intelligence Lab

Sep 2025 - Dec 2025

- Conducted research on **StyleGAN** and **Stable Diffusion** using **PyTorch** on an **HPC cluster**, managing the full **software development lifecycle** for generative models and conducting fine-tuning using **CUDA** and **LoRA**
- Architected model inference services using **vLLM** and **Hugging Face**, implementing **daemon processes** to handle high-concurrency requests and ensuring secure, efficient **visualization** and analysis of generated data
- Led workshops on **GPU programming** and building **AI Agents** for **50+ students**, focusing on technical communication and breaking down complex **architectural stacks** for a technical audience

Software Engineer

Noida, India

Qualitest

Feb 2021 - Dec 2023

- Designed and developed a cloud-agnostic **Storage Service** using **Python**, **PostgreSQL**, and **Redis**, architecting **APIs** that handle high-volume file operations across **AWS S3** and **Google Cloud Storage** via expiring signed URLs
- Contributed to an **Identity & Access Management** platform using **Java**, **Spring Boot**, and **AWS**, supporting thousands of users across **30+ applications** while adhering to **Agile principles** and participating in **stand ups** and **planning**
- Developed backend workflows using **AWS Step Functions** to process long-running **data warehouse** import jobs, preventing **Lambda** timeouts and optimizing **daemon process** execution for high-availability systems
- Maintained internal support tools using **LangChain** and **RAG**, reducing ticket resolution time by **20%** by enabling natural language queries over documentation and improving **developer productivity**

PROJECTS

Dark Factory - Orchestrator for AI coding CLIs | *Crossterm, Ratatui, Y-crdt*

github.com/ozzyozbourne/dark-factory

- Built an **event-driven orchestrator** for AI coding CLIs, enabling secure, real-time command execution and logging via **PTY** to prevent unauthorized bash shell access and ensure **system self-governance**

- Implemented a **distributed state** management system using **Y-crdt**, enabling reliable coordination across multiple machines and eliminating polling overhead through high-performance agentic loops